



AI-DRIVEN

CYBER SECURITY

WHAT IF YOUR AI COULD SEARCH THOUSANDS OF SECURITY DOCUMENTS, FIND THE RIGHT CVE, UNDERSTAND THE ATTACK TECHNIQUE, AND ANSWER YOU WITH EVIDENCE?

RAG

- COMBINES SEARCH + AI GENERATION
- RETRIEVES RELEVANT INFORMATION
- GIVES CONTEXT TO THE LLM
- REDUCES HALLUCINATIONS
- USEFUL FOR SECURITY KNOWLEDGE BASES

RAG WORKFLOW

- USER ASKS A QUESTION
- SEARCH KNOWLEDGE BASE
- RETRIEVE RELEVANT CHUNKS
- SEND CONTEXT TO LLM
- GENERATE FINAL ANSWER

CVE

- COMMON VULNERABILITIES AND EXPOSURES
- UNIQUE ID FOR A VULNERABILITY
- EXAMPLE: CVE-2024-XXXX
- DESCRIBES KNOWN SECURITY VULNERABILITIES
- HELPS TRACK AND RESEARCH VULNERABILITIES

CVE INFORMATION

- VULNERABILITY DESCRIPTION
- AFFECTED SOFTWARE
- SEVERITY
- REFERENCES
- REMEDIATION INFORMATION

MITRE ATT&CK

- KNOWLEDGE BASE OF ADVERSARY BEHAVIOR
- DOCUMENTS REAL-WORLD ATTACK TECHNIQUES
- USED BY SECURITY TEAMS
- HELPS DETECT AND INVESTIGATE ATTACKS
- SUPPORTS THREAT MODELING

TACTICS VS TECHNIQUES

- TACTIC: ATTACKER'S GOAL
- TECHNIQUE: HOW THE GOAL IS ACHIEVED
- EXAMPLE TACTIC: CREDENTIAL ACCESS
- EXAMPLE TECHNIQUE: BRUTE FORCE
- TECHNIQUES HAVE UNIQUE IDS

RAG + CVE + MITRE

- CVES → VULNERABILITY KNOWLEDGE
- MITRE → ATTACKER BEHAVIOR
- RAG → RETRIEVES RELEVANT INFORMATION
- LLM → EXPLAINS THE INFORMATION
- TOGETHER → AI SECURITY ASSISTANT